

EMMANUEL GODFREY CHUWA

Full-Stack Developer · Backend Engineer · Network Engineer

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PROFESSIONAL PROFILE

Final-year Software Engineering student at the University of Dodoma (UDOM) with hands-on full-stack experience across React.js, Vue.js 3, Node.js, Django, Laravel, and PHP. Delivered a centralised Single Sign-On (SSO) authentication system for UDOM integrating four institutional subsystems with secure JWT/OAuth 2.0 token logic, a full-stack school management platform currently in production use, and a live apartments booking website for a paying client in Dodoma. Comfortable working across relational and non-relational databases, RESTful API design, and system architecture. Coursework in Artificial Intelligence and Big Data Analytics has built a working foundation in algorithms, statistics, and mathematical modelling. Backed by practical field exposure at TANESCO and Huawei DSM. A fast learner who adapts quickly to new languages and stacks.

EDUCATION

2022 – 2026	Bachelor of Science in Software Engineering <i>University of Dodoma (UDOM), Tanzania</i>	-
2018 – 2021	Advanced Certificate of Secondary Education Examination (ACSEE) Tabora Teacher college	
2015 – 2018	Certificate of Secondary Education Examination (CSEE) <i>Kigogo Secondary School</i>	
2008 – 2014	Certificate of Primary Education <i>Kawawa Primary School</i>	

WORK EXPERIENCE

2024 – 2025	Huawei DSM <i>Field Attachment - IT Department / AUTIN Team</i> - Contributed to automation workflows within the AUTIN team, reducing manual operational tasks and improving team efficiency. - Assisted in the configuration and management of enterprise IT infrastructure, supporting stable uptime across systems. - Participated in deployment and maintenance of network and software systems serving multiple enterprise users. - Collaborated with cross-functional technical teams to troubleshoot and resolve infrastructure issues within agreed response times. - Produced technical documentation and project progress reports, improving knowledge transfer within the team.	-
2023 – 2024	TANESCO DSM <i>Field Attachment - IT Department / Networking</i> - Installed and configured routers, switches, and wireless access points, expanding network coverage across designated areas. - Diagnosed and resolved LAN/WAN faults, minimising network downtime and maintaining service continuity. - Executed structured cabling and cable management for multiple network distribution points, ensuring clean and documented infrastructure. - Assisted in IP address planning and subnetting, contributing to an organised and scalable network addressing scheme.	

- Supported IT staff in real-time monitoring of internal network systems, contributing to proactive issue identification.

KEY PROJECTS

UDOM Single Sign-On (SSO) System

Stack: Node.js · Express.js · React.js · PostgreSQL · JWT · OAuth 2.0 · OpenID Connect · bcrypt · Nodemailer

GitHub: github.com/echuwa

- Architected and developed a centralised SSO authentication platform integrating four UDOM subsystems (Student Records, IPT, LMS, E-Classroom), eliminating the need for separate logins per system.
- Implemented secure JWT-based authentication with OAuth 2.0 and OpenID Connect, ensuring token integrity and role-based access control across all integrated platforms.
- Designed a normalised PostgreSQL schema supporting unified identity management, system ID mapping via JWT claims, and migration history tracking.
- Produced comprehensive technical documentation including ERDs, use case diagrams, sequence diagrams, and system architecture documents meeting supervisor review requirements.
- Managed a multi-subsystem data migration strategy with temporary credential generation and migration audit trails.

John Merlin Secondary School Management System (JM-SMS)

Stack: Vue.js 3 · Django · MySQL · REST API

GitHub: github.com/echuwa

- Built a full-stack school management system handling student records, academic reporting, and staff management digitising previously manual administrative processes.
- Designed and implemented a RESTful backend API with Django, consumed by a responsive Vue.js 3 frontend, enabling real-time data access for school staff.
- Delivered the full development cycle from requirements gathering through to deployment and testing on Ubuntu Linux.

John Merlin Secondary School Website

Stack: HTML5 · Tailwind CSS · JavaScript

- Designed and developed a responsive institutional website with a modern UI, establishing the school's official digital presence.
- Managed social media account setup and branding across Instagram, Facebook, and TikTok (@johnmerlin_secondary_tz), growing institutional online visibility.

2Jem Apartments — Apartments & Hotel Booking Website (Dodoma)

Stack: Django · REST API · HTML5 · CSS3 · JavaScript

GitHub (Frontend): github.com/echuwa/2Jem_FE

GitHub (Backend): github.com/echuwa/2Jem_BE

Live: "http://jemapartments.co.tz"

- Delivered a live apartments and hotel booking website for a real client in Dodoma, currently in production at jemapartments.co.tz.
- Built the backend with Django and a REST API, paired with a custom HTML5, CSS3, and JavaScript frontend built to the client's specific requirements.
- Currently integrating a payment gateway API, pending credentials from the client, to enable direct online booking payments.

TECHNICAL SKILLS

Programming Languages	JavaScript (ES6+), Python, PHP, Java
Frontend Development	Vue.js 3, React.js, HTML5, CSS3, Tailwind CSS
Backend Development	Node.js, Express.js, Django, Laravel, PHP, worked across multiple frameworks and languages, not Python alone
Databases	PostgreSQL, MySQL, MongoDB

Authentication	JWT, OAuth 2.0, OpenID Connect, bcrypt
Networking	LAN/WAN Configuration, Troubleshooting, Structured Cabling, IP Planning, Subnetting, Active Directory, Group Policy (GPO), DNS, DHCP, WDS
DevOps & Tools	Git, GitHub, GitHub Actions, Linux (Ubuntu), REST APIs
Math & Algorithms	Search algorithms, CSPs, regression & classification models, neural networks/CNNs (AI & Big Data Analytics coursework)

HOBBIES & INTERESTS

Open-source contribution · Technical writing & blogging · Community development · Photography · Sports · Learning new technologies

REFEREES

1. Salvator Gaspar Chuwa <i>Head of Investigation Department</i> Fair Competition Commission (FCC), Dodoma ✉ salvator.chuwa@fcc.go.tz ☎ +255 784 306 594	3. Ms. Rose George Lugano Tutorial Assistant University of Dodoma (UDOM) ✉ georgerose1431@gmail.com ☎ +255 782 673 208 / +255 762 728014
2. Mr. Erick Samwel Ndalohwa Tanzania AUTIN – Huawei Lead ✉ erick.samwel.ndalohwa2@huawei.co.tz ☎ +255 748 173 751	4. Mr. Emmanuel M Dato Maritime Transport Officer ✉ emxymichaelb@yahoo.com ☎ +255 716 479 219

I, Emmanuel G. Chuwa, certify that all information contained here in is true, complete, and accurately represents my qualifications and experience.